ORBCOMM®

WHERE DATA DRIVES DECISIONS



ST 21xx

Installation Guide

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CONTACT AND LEGAL INFORMATION

Visit ORBCOMM Online

www.ORBCOMM.com

Contact Support

Website:	https://partner-support.orbcomm.com
Email	partner-support@orbcomm.com
Phone	+1.613.836.2222

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ABOUT THIS GUIDE

IMPORTANT

READ ALL INSTRUCTIONS CAREFULLY BEFORE INSTALLING. FAILURE TO DO SO MAY CAUSE PERSONAL INJURY OR DAMAGE TO PRODUCT AND/OR PROPERTY.

• Review the product package and contents prior to beginning the installation. Take care when opening the packaging and removing items. If a return is needed, you will want to return the product in its original packaging if possible. • This instruction guide is provided as a GENERAL installation guide; some assets vary dimensionally and may require additional steps. • ORBCOMM has a policy of continuous development and improvements. Therefore, products, guides, and technical information are subject to change without prior notice. • The manufacturer and / or distributors do not accept responsibility for third-party charges, labor, and / or third-party replacement modifications that are not ORBCOMM approved. Some modifications may void the factory warranty. • ORBCOMM does not accept any responsibility for installations performed by installers / third parties not approved and / or authorized by ORBCOMM. Some installations may void the factory warranty.• Exercise due diligence when installing this product. ORBCOMM does not accept any responsibility for asset damage or personal injury resulting from the installation of this product. Careless installation and operation can result in serious injury or equipment damage. • All liability for installation and use rests with the owner / operator. • Always make sure you have a clean, dry, and well-lit work area. • Always ensure products are secure during disassembly and installation. • Always take steps to protect yourself when drilling, cutting, and grinding because this may create flying particles that can cause injury. • Thoroughly inspect the area to be drilled, on both sides of material, prior to modification, and relocate any objects that may become damaged. • Always route electrical cables carefully. Avoid moving parts, parts that may become hot and rough, or sharp edges. • Make sure to fully understand the product, its intended use, and operation prior to use.

Purpose

This guide contains product information for the ST 21xx. The intended audiences for this guide include field support personnel, product evaluators, and certified third-party personnel. It is particularly intended for personnel who are responsible for system installation and activation. In addition, and as is appropriate, this guide may be used for training purposes.

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OVERVIEW

The ST 21xx is manufactured by ORBCOMM and can be used for side or bottom mount installations, either using VHB tape or screws.

This guide provides information required for a successful, reliable installation of the ST 21xx on a vehicle, boat, or other platform. This guide does not address the ST 21xx commissioning procedures as these procedures vary depending upon the Solution Provider and their specific application.

Figure 1: ST 21xx



The following outlines the overall procedure for installing and activating an ST 21xx.

1. Gather the required tools and materials.
2. Record the mobile ID.
3. Review the mounting guidelines.
4. Install the ST 21xx.
    - Cab mount installation
    - Surface mount installation
        - Screw mount
        - VHB mount
5. Apply power and register the ST 21xx.

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1 GATHER THE REQUIRED TOOLS AND COMPONENTS

CAUTION: While ORBCOMM provides mounting hardware to assist with installations, it is the responsibility of the installer to select the proper mounting hardware for the asset's surface material where an ORBCOMM device or accessory will be mounted.

The following components are available from ORBCOMM:

- An ST 21xx (includes cable/pigtail)
- A mounting kit (one of the following)
    - Cab Mount Kit (ST100990-001):

      Overmolded mounting plate (Qty 1)

      Counter sunk screws (Qty 4)

      Modified locking nut (Qty 1)

  OR

    - Surface Mount Kit (for screw or double-sided tape installations) (ST100991-001):

      Mounting plate (Qty 1)

      Counter sunk screws (Qty 4)

      Hex head drilling screws (Qty 4)

      Silicone plug (Qty 1)

      Surface preparation material (surface preparation pad, alcohol wipes, primer)

  OR

    - Magnet Mount Kit (ST101419-001)

      Mounting plate (Qty 1)

      Counter sunk screws (Qty 4)

      Magnet kit (magnet, magnet pad, screw, and lock nut) (Qty 4 of each)

      Silicone plug (Qty 1)

      Surface preparation material (surface preparation pad, alcohol wipes)

The following tools and materials are required (customer supplied) for a non-magnet mount installation:

- Drill and 16 mm or 5/8" drill bit (cab mount installations only)
- Silicone caulking (cab mount installation only)

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2 RECORD THE MOBILE ID

Each ST 21xx has a unique serial number, termed a mobile ID, used by ORBCOMM to register it on the IsatData Pro network. This is a 15-digit alphanumeric identifier in the format "NNNNNNNNSKYXXXX" The mobile ID is located on the bottom of the ST 21xx and on the shipping box.

1. Record this mobile ID before the ST 21xx is installed. You need it later to activate the ST 21xx on the network.

Note: The ST 21xx might be activated on the network prior to or after shipping based on the Solution Provider agreement.

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3 REVIEW THE MOUNTING GUIDELINES

CAUTION: Most users install the ST 21xx on a vehicle. It is very important for installers to install the ST 21xx in a safe and secure way to avoid danger or damage to persons or property.

- Mount the ST 21xx where it has a clear view of the sky/satellite. For a mobile installation, this means that it is preferable to install at the highest point on the vehicle or vessel where it has a clear view of the sky in all directions.
- Mount the ST 21xx on a flat surface for mobile installations such that the elevation angle does not change with rotation.
- Mount on a surface that is clean and free of oil, dirt, and debris.
- Fasten the ST 21xx securely so that it is not loose and does not move easily.
- Mount the ST 21xx on a solid, stable surface.
- Ensure that any paint above the ST 21xx is non-metallic and non-metallic flake, if the installation is under fiberglass or composite wind fairings.
- Check that the ST 21xx's cable reaches the power source before you drill any mounting holes.
- Mount the ST 21xx on the driver's side of the vehicle, if possible, when there is a possibility of strikes by overhanging tree branches.
- Mount the ST 21xx on a surface that does not get hotter than the maximum operating temperature (+85°C/+185°F). If the surface may get hotter, mount the ST 21xx with a thermal barrier between it and the mounting surface.

CAUTION: Mount the ST 21xx at least 20 cm (8 in.) away from humans.

- DO NOT mount the ST 21xx close to other electrical equipment due to possible radiated and/or conducted electromagnetic interference.
- DO NOT mount the ST 21xx close to radar or other communications antennas. Use the following guidelines:

  > 1 m from VHF/UHF antenna

  > 3 m from loop antenna

  > 4 m from MF/HF antenna

  > 5 m from other satellite antennas

  Not within a radar beam
- DO NOT mount the ST 21xx where water may build-up or collect.
- DO NOT mount the ST 21xx close to an exhaust pipe due to the excessive heat and the potential for the exhaust pipe causing satellite blockage.
- DO NOT mount the ST 21xx close to air horns or any tractor roof hardware (for example, emergency lights) that could interfere with satellite communications.
- DO NOT install the ST 21xx inside the truck under the roof liner.

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4 INSTALL THE ST 21XX

1. Once you have identified a mounting location, determine the type of installation required, and then follow the instructions for that type of installation.
    - Cab mount (Section 4.1)
    - Surface mount (Section 4.2)
        - Screw mount (Section 4.2.1)
        - Double-sided tape mount (Section 4.2.2)
    - Magnet mount (Section 4.3)

4.1 Cab Mount Installation

Note: The cab mount installation is rated IPx6 for Pressure, as per IEC 60529, when installed following the instructions in this guide.

1. Apply a bead of silicone caulking around the top side of the internal flange that surrounds the threaded shank, as per the figure.

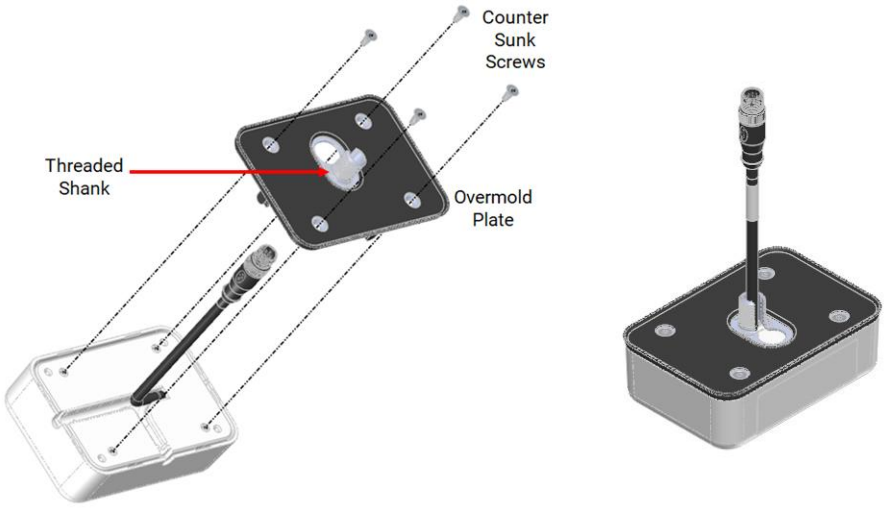


2. Pass the ST 21xx pigtail through the threaded shank on the overmolded plate, and then securely fit it into place.



3. Assemble the overmolded plate onto the ST 21xx with the four (4) counter sunk screws using 1 N-m (8.5 in-lb) of torque. The plug features on the plate line up with the cable channels

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4. Add a second bead of caulking / RTV around the sealing lip (shown in red in the figure).



5. At the mounting location, drill a 16 mm (5/8”) diameter hole in the asset, and then pass the pigtail through the hole in the asset.
6. Install the device to the asset.
7. On the inside of the cab, assemble the modified lock nut, and then torque to a minimum of 2.25 N-m (20 in-lbs).



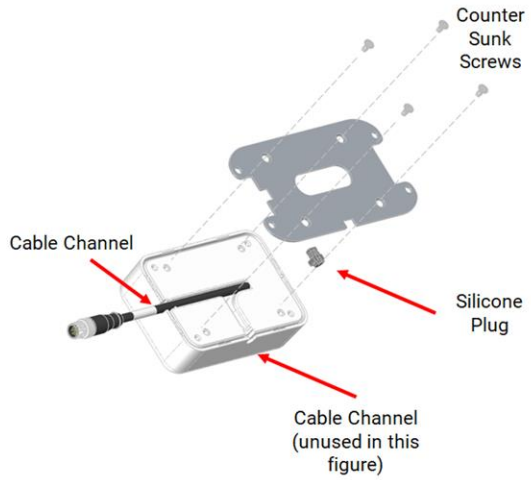
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4.2 Surface Mount Installation

Two surface mount options are available: screw or double-sided tape.

4.2.1 Screw Mount Installation

1. Use the surface mounting plate as a template to mark the installation location for the ST 21xx. Ensure that the hole positions are marked.
2. Route the pigtail into one of the two cable channels.
3. Install the silicone plug into whichever cable channel is not used.



4. Assemble the mounting plate onto the ST 21xx using the four (4) counter sunk screws (see figure above) using 1 N-m (8.5 in-lb) of torque.



5. Position the assembled ST 21xx on the marked installation location, and then use the four (4) hex head drilling screws to affix the ST 21xx to the asset.

   CAUTION: DO NOT use the double-sided tape on the mounting plate.

   CAUTION: DO NOT over-troque the screws.

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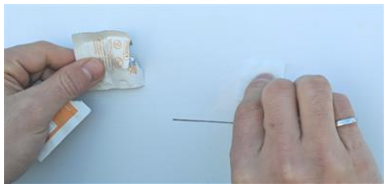


4.2.2 Double-Sided Tape Mount Installation

1. Read the important inclement weather guidelines in APPENDIX B before starting the installation.
2. Ensure that the mounting surface and the double-sided tape are both at least 15°C (60°F). If necessary, use a heat lamp or heat gun to warm them.
3. Use the supplied surface preparation pad to lightly scratch the surface of the asset at the mounting location.



4. Use the supplied alcohol pad to clean the mounting surface. Ensure that the area is clear of any dirt or particles. Clear the surface with an alcohol cleaner or wipe.

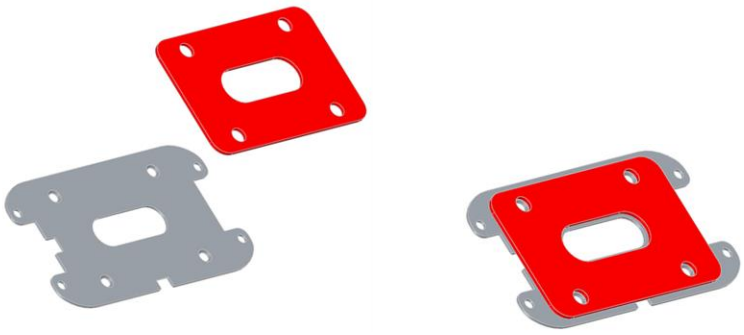


5. Squeeze the supplied primer tube to crush the insert and release the primer, and then apply the primer to the mounting surface. Apply the primer to the area, and then wait at least 30 seconds for the primer to dry.

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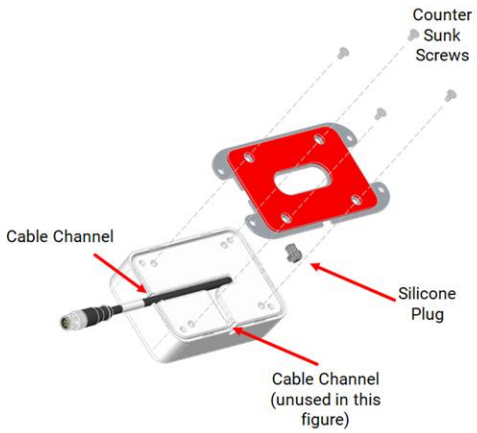
6. Use the surface mounting plate as a template to mark the installation location for the ST 21xx.
7. Remove one of the liners from the double-sided tape pad, and then affix it to the cleaned mounting plate ensuring that the screw holes on the two surfaces line up and air bubbles are minimized.



CAUTION: Do not touch the tape.

CAUTION: Double-sided tape installation temperature must be above 15°C (60°F). Refer to APPENDIX B.

8. Route the ST 21xx pigtail into one of the two cable channels.
9. Install the silicone plug into whichever cable channel is not used.



10. Assemble the mounting plate / tape pad onto the ST 21xx using the four (4) counter sunk screws (see figure above).

CAUTION: Do not overtroque the screws.

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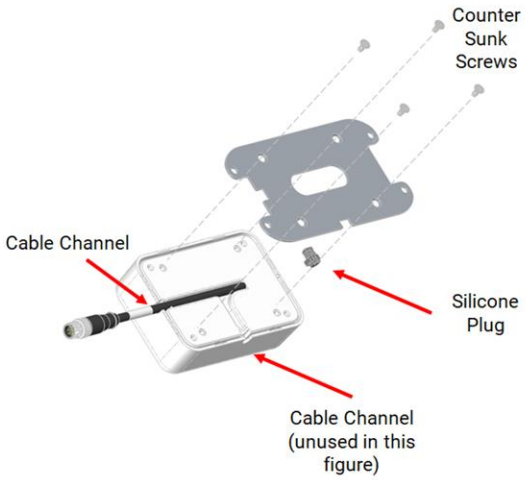
11. Carefully remove the second liner to assemble the ST 21xx to the asset.
12. Position the assembled ST 21xx on the marked installation location.



13. Press firmly on the entire top surface of the ST 21xx (15 pounds for 60 seconds) to bond the double-sided tape to the asset.

4.3 Magnet Mount Installation

1. Route the pigtail into one of the two cable channels.
2. Install the silicone plug into whichever cable channel is not used.



3. Assemble the mounting plate onto the ST 21xx using the four (4) counter sunk screws (see figure above) using 1 N-m (8.5 in-lbs) of torque.

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4. Assemble the magnetic feet onto the mounting plate using the four (4) sets of magnet, magnet pad, Phillips screw, and lock nut (see figure above).

   CAUTION: DO NOT over-torque the screws.

5. Use the provided surface preparation kit to clean the mounting surface before applying the magnet pad to the asset surface.
6. Position the assembled ST 21xx on a ferromagnetic surface on the asset.

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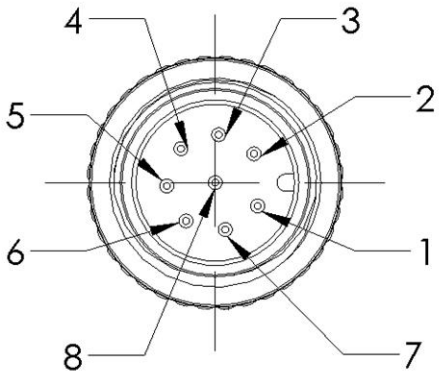
5 APPLY POWER AND REGISTER THE ST 21XX

When you apply power, the ST 21xx goes into satellite search mode to acquire the IsatData Pro network. This activity may take a few minutes to complete. The ST 21xx must complete registration to operate.

Once the ST 21xx synchronizes itself with the network, it sends a registration message to the network. The ST 21xx will not register until it has a clear line-of-sight to the satellite. The network records the registration message and forwards the registration message to the user's application. The network sends an acknowledgment message over the satellite to the ST 21xx. The ST 21xx is now available to send and receive messages via satellite.

1. Before connecting the ST 21xx to an external voltage source, ensure that the polarity is correct, and the voltage source is between 9 and 32 VDC. Refer to the connector pin assignment description.

Figure 2: ST 21xx Pin Assignment



PIN	Description
1	External Reset Input
2	Ground
3	RS-232 Rx (Input)
4	RS-232 Tx (Output)
5	VEXT
6	1 PPS Output
7	Event Notification Output
8	External Reset Output

2. Connect the ground connection at the same time or before applying power.
3. Check the ST 21xx's LED to ensure it has successfully powered on. The visual indicator (LED) does not indicate satellite status. Its only purpose is to confirm the ST 21xx has power.

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APPENDIX A TROUBLESHOOTING

The following section contains troubleshooting information.

Situation : ST 21xx Does Not Register or Report

On application of external power, the ST 21xx should register with the satellite and send a report.

Note: The ST 21xx must be visible to the satellite to register with the network and registration normally takes a few minutes.

If the ST 21xx fails to register or report:

- Ensure that the ST 21xx has a clear line-of-sight to the satellite. The ST 21xx must operate outdoors and be unobstructed by buildings, forest canopy, and rock cuts.
- Check that no objects or debris are on the ST 21xx and blocking transmission.
- Verify with your supplier that the ST 21xx is assigned to your account and registered (that it is sending and receiving) and that the IsatData Pro network is operating properly.

Replace the ST 21xx if the earlier checks fail to uncover the problem.

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APPENDIX B INCLEMENT WEATHER GUIDELINES

The ST 21xx can be secured to an asset using double-sided tape. Proper double-sided tape application requires that the tape is kept warm, and the asset surface is both clean and dry.

CAUTION: Failing to follow these guidelines will compromise the installation.

Mandatory Guidelines for Installation in Wet Weather

The installation surface on the asset must be completely clean and dry for the tape to bond. If it is raining or snowing hard enough that the surface cannot be kept dry, do NOT proceed with the installation.

Mandatory Guidelines for Installation in Cold Weather

Below (15°C (60°F) the tape starts becoming firm which makes it more difficult to bond to the asset.

If the guidelines below are followed correctly the ST 21xx can be installed at temperatures down to -20°C (-5°F).

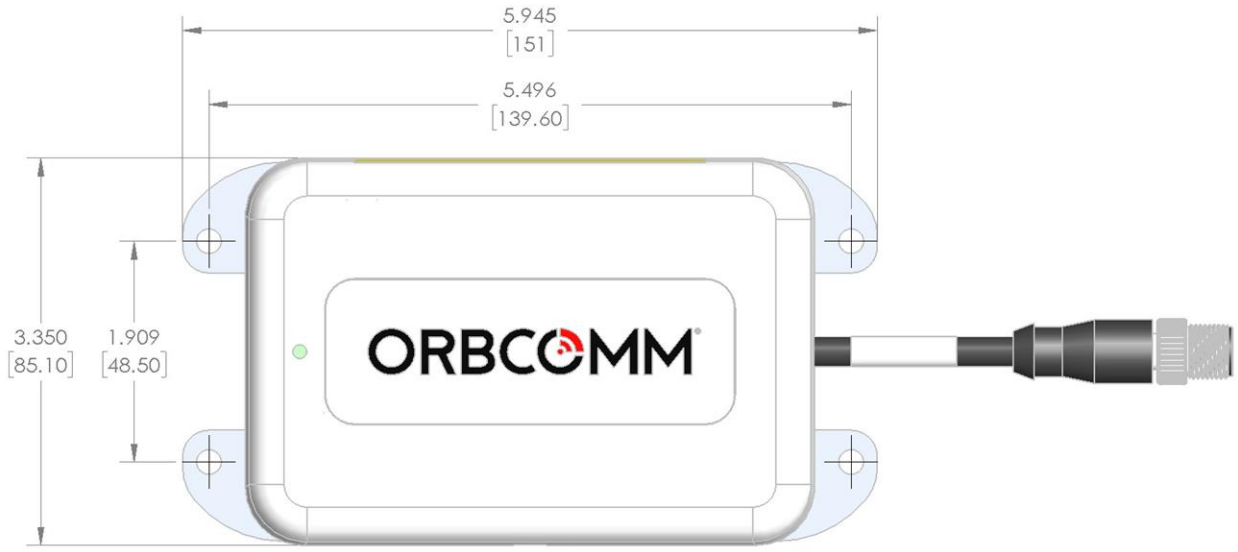
CAUTION: The double-sided tape mounting brackets must not be used at temperatures below -20°C (-5°F).

- At or below freezing temperatures (0°C or 32°F), both the ST 21xx and the primer must be kept at room temperature, for example, inside an idling vehicle or a warm building.
- Keep the primer warm (room temperature) until ready to apply. The primer will not dry quickly at cold temperatures however, in this situation the tape should be applied when the primer is still wet, as it improves initial bond.
- Keep the ST 21xx warm (room temperature) until it is time to mount it to the asset.
- Press firmly on the entire top surface of the ST 21xx (15 pounds for 60 seconds) to bond the tape to the asset.

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APPENDIX C SURFACE MOUNTING PLATE DIMENSIONS

The dimensions for the surface mounting plate are shown below.

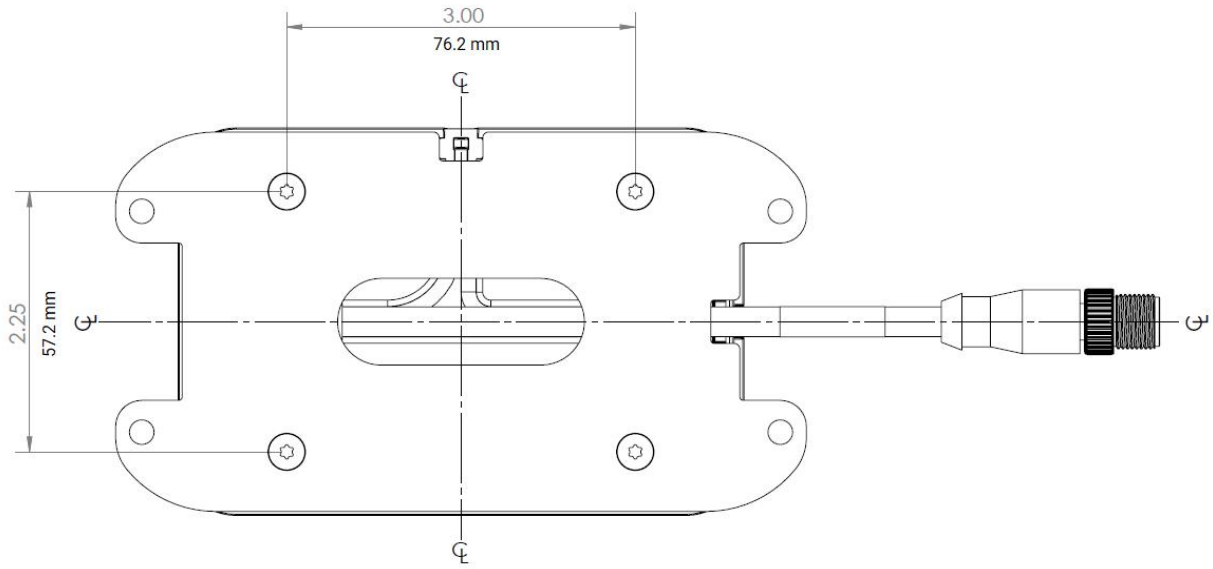


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APPENDIX D ST 21XX PLATE HOLES

CAUTION: DO NOT USE THIS FIGURE AS A TEMPLATE - IT IS NOT TO SCALE. Use the physical ST 21xx hardware as a template if necessary.

CAUTION: ORBCOMM is not responsible for issues with or caused by assembles or installations using custom plates not manufactured by ORBCOMM.



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